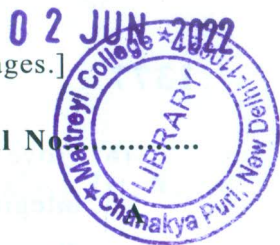


[This question paper contains 8 printed pages.]



Your Roll No.

Sr. No. of Question Paper : 1377

Unique Paper Code : 32351402

Name of the Paper : BMATH-409; Riemann
Integration and Series of
Functions

Name of the Course : **B.Sc. (H) Mathematics**

Semester : IV

Duration : 3 hours + 30 minutes Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Attempt any **two** parts from each question.
3. **All** questions are compulsory.

1. (a) Let f be a bounded function on $[a, b]$. Define integrability of f on $[a, b]$ in the sense of Riemann.

(6)

P.T.O.

- (b) Prove that every continuous function on $[a, b]$ is integrable. Discuss about the integrability of discontinuous functions. (6)

- (c) Let $f(x) = \sin \frac{1}{x}$ for $x \neq 0$ and $f(0) = 0$. Show that

$$f \text{ is integrable on } [-1, 1], \text{ Show that } \left| \int_{-1}^1 f(t) dt \right| \leq 2. \quad (6)$$

- (d) Let $f(x) = x$ for rational x ; and $f(x) = 0$ for irrational x . Calculate the upper and lower Darboux integrals of f on the interval $[0, b]$. Is f integrable on $[0, b]$? (6)

2. (a) State Fundamental Theorem of Calculus II. Use it to calculate

$$\lim_{x \rightarrow 0} \frac{1}{x} \int_0^x e^{t^2} dt. \quad (6.5)$$

- (b) Let f be defined as (6.5)

$$f(t) = \begin{cases} 0, & t < 0 \\ t, & 0 \leq t \leq 1 \\ 4, & t > 1 \end{cases}$$

- (i) Determine the function $F(x) = \int_0^x f(t) dt$.

- (ii) Sketch F . Where is F continuous?

- (iii) Where is F differentiable? Calculate F' at points of differentiability.

- (c) State and prove Intermediate value theorem for Integral Calculus. Give an example to show that condition of continuity of the function cannot be relaxed. (6.5)

- (d) Let f be a continuous function on $\mathbb{R}$. Define

$$G(x) = \int_0^{\sin x} f(t) dt \text{ for } x \in \mathbb{R}.$$

- Show that G is differentiable on $\mathbb{R}$ and compute G' . (6.5)

3. (a) Let $\beta(p, q)$ (where $p, q > 0$) denotes the beta function, show that

$$\beta(p, q) = \int_0^1 \frac{u^{p-1}}{(1+u)^{p+q}} du = \int_0^1 \frac{v^{p-1} + v^{q-1}}{(1+v)^{p+q}} dv. \quad (6)$$

- (b) Determine the convergence and divergence of the following improper integrals

(i) $\int_0^1 \frac{dx}{x(\ln x)^2}$

(ii) $\int_{-1}^{\infty} \frac{dx}{x^2 + 4x + 6}$ (6)

- (c) Define Improper Integral of type II.

Show that the improper integral $\int_1^{\infty} \frac{dx}{x^p}$ converges iff $p > 1$. (6)

- (d) Show that the improper integral $\int_{\pi}^{\infty} \frac{\sin x}{x} dx$ is convergent but doesn't converge absolutely. (6)

4. (a) Let $\langle f_n \rangle$ be a sequence of integrable functions on $[a, b]$ and suppose that $\langle f_n \rangle$ converges uniformly on $[a, b]$ to f . Show that f is integrable. (6)

- (b) Define

(i) pointwise convergence of sequence of functions

(ii) uniform convergence of a sequence of functions

(iii) If $A \subseteq \mathbb{R}$ and $\emptyset: A \rightarrow \mathbb{R}$ then define uniform norm of $\emptyset$ on A . (6)

(c) (i) Discuss the pointwise and uniform convergence of $f_n(x) = \frac{x}{n}$ for $x \in \mathbb{R}$, $n \in \mathbb{N}$.

(ii) Show that the sequence $\langle f_n \rangle$ where $f_n(x) = \frac{n}{x+n}$, $x \geq 0$ is uniformly convergent in any finite interval. (6)

(d) (i) Show that the sequence $\langle f_n \rangle$ where $f_n(x) = \frac{\sin nx}{\sqrt{n}}$ uniformly convergent on $[0, \pi]$.

(ii) Discuss the pointwise and uniform convergence of the sequence $g_n(x) = x^n$ for $x \in \mathbb{R}$, $n \in \mathbb{N}$. (6)

5. (a) (i) State and prove the Cauchy Criteria for uniform convergence of series. (3.5)

- (ii) Show that the series $\sum_0^\infty (1-x)x^n$ is not uniformly convergent on $[0, 1]$ (3)

- (b) Show that $\sum \frac{(-1)^n}{n^p} \frac{x^{2n}}{(1+x^{2n})}$ converges absolutely

and uniformly for all values of x if $p > 1$.

- (c) Is the sequence $\langle f_n \rangle$ where $f_n = \frac{\sin(nx+n)}{n}$, uniformly convergent on $\mathbb{R}$? Justify. (6.5)

- (d) If f_n is continuous on $D \subseteq \mathbb{R}$ to $\mathbb{R}$ for each $n \in \mathbb{N}$ and $\sum f_n$ converges to f uniformly on D then prove that f is continuous on D . (6.5)

6. (a) Find the radius of convergence and exact interval of convergence of the following power series :

(i) $\sum_{n=1}^\infty \frac{(-1)^{n+1}}{n^2} (x+1)^n$

(ii) $\sum_{n=0}^\infty x^{n!}$ (6.5)

- (b) Let $f(x) = \sum_{n=0}^\infty a_n x^n$ has radius of convergence $R > 0$. Show that the function f is differentiable on $(-R, R)$ and

$f'(x) = \sum_{n=1}^\infty n a_n x^{n-1}$ for $|x| < R$. (6.5)

- (c) Show that

(i) $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$

for $|x| < 1$

(ii) $\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots$. (6.5)

- (d) Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence $R > 0$. If $0 < R_1 < R$, show that the power series converges uniformly on $[-R_1, R_1]$. Also, show that the sum function $f(x)$ is continuous on the interval $(-R, R)$. (6.5)